Low and Negligible Risk Research Project Description

HUMAN RESEARCH ETHICS

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Purpose

- The Project Description is a mandatory component of submission of the Human Research Ethics Application (HREA) form. It is uploaded as part of your HREA through the UQ MyResearch system.
- The Project Description provides the background, rationale and methodological approach of a research project being submitted for ethics review.
- This template has been developed specifically to meet the needs of ethics review processes at UQ for Low and Negligible Risk (LNR) research. The information contained in this form and the HREA will assist reviewers in assessing the application against the principles outlined in the *National Statement on Ethical Conduct in Human* Research (2007, updated 2018), namely:
    - Research Merit and Integrity
    - Justice
    - Beneficence
    - Respect
- There is no need to duplicate information in the HREA into the Project Description or vice versa. It is recommended that you familiarise yourself with requirements of the MyResearch HREA so that information is not inadvertently duplicated in this Project Description. For example, for LNR research, information related consent, recruitment, risks and benefits is addressed in the HREA and does not need to be duplicated in the Project Description.
- To assist in completing this template, examples are provided of the type of information required in each section.
- In completing this form please:
    - Use plain/lay language, defining all technical terms and avoiding acronyms.
    - Keep to the section limits without altering font size (Calibri 11 pt) or margins (2.5 cm). Each section (questions 2-4) should begin on a separate page.
    - Be as concise as possible; it is not necessary to exhaust the page limits.

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1. Project Details	• As listed in your HREA form in MyResearch
A. Title	Measuring Episodic Memory with Virtual Reality
B. MyResearch application #	2022/HE001298
C. Lead/Chief Investigator	Professor Jason Mattingley
D. Project Description version/date	

2. Research Scope, Aims, Themes or Questions (maximum one page)	• What is the research theme or question that this project is designed to explore? (NS 3.1) • Why is the exploration of this theme or answer to this question worth pursuing? (NS 3.1)

A. Research theme or question

Research theme: Validation of a new virtual reality episodic memory metric.

B. Background and specific aims

Episodic memory relates to the "what, when, and where" of recent experiences. Remembering what we had for dinner last night, where we left our car keys, or the name of a new person in the office, are all examples of episodic memories. Researchers are interested in measuring episodic memory to advance basic scientific understanding, and because impairments of episodic memory are a common symptom of brain injury and disease (e.g., stroke, dementia).

Measuring a person's episodic memory is difficult. Typically, lists of words are read aloud to a person and their word recall ability is assumed to be equivalent to episodic memory performance.

Modern memory research shows, however, that performance across memory domains, i.e., verbal, spatial, episodic, procedural etc. can be independent of each other. Performance in one domain isn't always predictive of performance in another.

The present study aims to establish the utility and validity of a new metric to directly assess episodic memory in normal adults. This will be achieved using a virtual reality (VR) task and follow-up questionnaires designed by the study team.

Four virtual environments are presented to participants and instructions are given to complete simple tasks within each one, as described in detail below. Participants are then asked questions about each environment via four online questionnaires. The first questionnaire is to be answered immediately after the participant's VR experience. The remaining three are to be completed over the course of several days or weeks.

By creating episodic memories via a virtual experience and then measuring recall accuracy over several weeks, it is anticipated that a more valid and accurate estimate of an individual's episodic memory function can be determined.

Validity will be established by having participants complete three standard cognitive tests of episodic memory, such as word-list learning, and calculating correlations between those results and the VR test results.

C. Significance

Valid measurement of episodic memory performance will allow neuropsychologists and physicians to make better informed clinical decisions and diagnoses in cases where people complain of episodic memory loss.

Furthermore, subjective episodic memory complaints that are too subtle to be detected by standard cognitive testing may be quantifiable via a test that directly measures episodic memory.

Finally, the ongoing search for an effective intervention to improve episodic memory could greatly benefit from the use of a metric that measures episodic memory directly. The present study's use of follow-up tests over the course of weeks could be used to inform future researchers of the efficacy of their intervention and how often treatments should be applied.

3. Research Methods (maximum two pages)	• How will the planned methods explore the theme or achieve the aims of the research? (NS 3.1) • Demonstrate that the research has merit (NS 1.1)

A. Research participants or data

One hundred participants aged 18+ years will be recruited. Eligibility criteria will include no known cognitive impairments, no past significant discomfort using virtual reality, and that the participant can climb a flight of stairs unassisted (required to access the VR testing space). These criteria will be assessed via participant self-report.

Participants will be sourced via The University of Queensland's Psychology Research Pool and through word of mouth.

Data will be collected and stored in two separate databases, identifying, and non-identifying.

Identifying information will include first name, last name, email address, phone number, and signature. This will be temporarily stored in a password protected format at https://REMEMVR.com.au prior to participation and destroyed after participation is complete.

Non-identifying information will include age, sex, education level, VR experience, history of brain injury, current medications, cognitive test/questionnaire answers, and VR test answers. This will be stored online in a password protected Google Sheets spreadsheet.

Individual participants in the non-identifying database will be labelled using numbers. No record of this number will be kept in the identifying database. This will prevent any possibility of linking a person's identifying information with their test results.

Signed consent forms will be scanned and uploaded to the UQ RDM with the hard copies to be destroyed afterwards.

Any data collection will occur only following a participant's explicit informed consent.

B. Methodology and timeline

Potential recruits will first visit https://REMEMVR.com.au, read information about the experiment, confirm they are eligible, and select a time to participate. This booking requires a participant to give their name, email address, and phone number. This information is used to send a confirmation of booking email and an SMS reminder on the day prior to participation and is deleted once participation is complete.

Participants arrive at QBI at their booked time and are escorted to the VR lab. They again read an information sheet explaining the experimental method and sign a consent form. Participants then complete a demographics and health information form. Then four standard cognitive tests and questionnaires are completed (Section D). This takes approximately 90 minutes.

Participants are then fitted with a sanitised VR headset (resembling a large eye-mask held in place by a comfortable headband) and are shown four virtual rooms of an imaginary house. In each room (e.g., kitchen, bathroom), participants are asked to describe various details such as the wall colour, position of furniture, or any objects that don't belong (e.g., a petrol can in the bathroom). Instructions are then given to participants to move six virtual objects around the room using reach and grasp actions. Once instructions for a room are completed, participants remove the VR headset and take a short break. Instructions for each room take approximately 10-15 minutes to complete, giving a total time in virtual reality of around 40-60 minutes.

Upon completion of all four rooms in the VR environment, participants use a computer to log in to https://REMEMVR.com.au to complete the first of four recall tests. One recall test takes

approximately 30 minutes. Once the first test is complete, participants may leave the lab. Total time for in-person participation at QBI is expected to be approximately three hours.

Participants complete a further three follow-up memory tests online via https://REMEMVR.com.au. Follow-up memory tests occur up to 30 days after the VR session. The timing of when each test is to be completed will not be identical for all participants and will be calculated individually to ensure an equal amount of data is collected over the 30-days across all participants. Once participants have completed the four recall tests online, they will be reimbursed for their time at $20 per hour (the standard hourly UQ rate). Total participation time for the entire study, including the initial VR sessions, is estimated to be five hours ($100).

Participants will receive payment for participation after the completion of their fourth REMEMVR test via direct bank transfer as per QBI Finance policy.

Data collection will be ongoing for the duration of the author's PhD.

C. Sampling strategy

A sample size of 100 participants has been deemed sufficient to establish preliminary normative data. Recruitment of healthy adults via the undergraduate pool and via word of mouth has been deemed appropriate to establish generalisable data. Additional recruitment will be made through the use of flyers on community noticeboards.

D. Measures and procedures

Copies of all measures (other than cognitive tests and the DASS) are attached to the HREA.

Demographic Questions: Age, sex, education level, VR experience.

Health Questions: VR sickness, head injuries, medications, pregnancy.

Questionnaires: Depression Anxiety and Stress Scale (DASS).

Cognitive Tests: National Adult Reading Test (NART), Rey Auditory Verbal Learning Test (RAVLT), Brief Visuospatial Memory Test Revised (BVMT-R), Raven's Progressive Matrices (RPM)

VR Memory Test: Recall test responses

E. Data analysis

To establish the VR task's criterion validity, correlational analysis will be conducted between the scores on cognitive tests, questionnaires, and the VR task's score.
Data will be reviewed regularly to check for sources of bias, ceiling/floor effects, and to allow for small changes to be made to the VR test itself. Any significant changes will be cleared with UQ Ethics first.

F. Other information related to research methods (where relevant)

Participants will be compensated at a rate of $20 per hour. Full participation is expected to take five hours. Compensation will be paid via direct bank transfer to participants after the completion of the fourth memory recall test.
During the consent process, participants will be informed of their right to withdraw from participation at any time without penalty or requiring an explanation.

G. Research merit

Should the research aims be met, the measurement of episodic memory via virtual reality will offer a more valid assessment of an individual's episodic memory, beyond current metrics. This could allow future research to explore its use as a tool for clinical diagnosis of memory impairment.

Furthermore, interventions that aim to improve an individual's episodic memory could employ this new virtual reality tool to measure small changes in an individual's episodic memory capacity across several memory-based theoretical domains. This could inform future researchers which domains of episodic memory have been modulated.

4. Other ethical issues (maximum half page)

- Dissemination of project outputs and outcomes (NS 3.1.69)
- Other ethical issues to your research project?

A. Research Outputs

The introduction, methods, results, and discussion of this research will be submitted for publication to a scientific journal.

Participants will be asked if they wish to be notified when data from this study are published in a journal article. If they wish to be notified, they will be asked to write their email in a dedicated password protected Excel document that contains only email addresses. Upon publication, a copy of the final journal article will be forwarded to each email address on the list.

B. Additional information

A participant information sheet, consent form, and demographic/health questionnaires are attached to the HREA in Q4.2